**Programiz**

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main.c

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```c
//BIN PACKING PROGRAM
#include <stdio.h>
#include <stdlib.h>

#define MAX_ITEMS 20

int n;
int capacity;
int weight[MAX_ITEMS];
int min_bins;
int bin_load[MAX_ITEMS];

// Sort in descending order
int compare(const void *a, const void *b) {
    return (*(int*)b - *(int*)a);
}

void get_initial_upper_bound() {
    int bins_used = 0;
    int temp_load[MAX_ITEMS] = {0};

    for (int i = 0; i < n; i++) {
        int placed = 0;
```

Output

Clear

```
--- Bin Packing: Branch and Bound ---
Enter number of items: 8
Enter bin capacity: 9
Enter weight of 8 items:

=== Session Ended. Please Run the code again ===
```

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```c
                placed = 1;
                break;
            }
        }
        if (!placed) {
            temp_load[bins_used] = weight[i];
            bins_used++;
        }
    }
    min_bins = bins_used;
}

void solve(int item_idx, int bins_used) {
    if (bins_used >= min_bins)
        return;

    if (item_idx == n) {
        if (bins_used < min_bins)
            min_bins = bins_used;
        return;
    }

    for (int i = 0; i < bins_used; i++) {
        if (bin_load[i] + weight[item_idx] <= capacity) {
            bin_load[i] += weight[item_idx];
            solve(item_idx + 1, bins_used);
```

Output

```
--- Bin Packing: Branch and Bound ---
Enter number of items: 8
Enter bin capacity: 9
Enter weight of 8 items:

=== Session Ended. Please Run the code again ===
```

**Programiz**

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main.c

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```c
    }
}

int main() {
    printf("--- Bin Packing: Branch and Bound ---\n");

    printf("Enter number of items: ");
    scanf("%d", &n);

    printf("Enter bin capacity: ");
    scanf("%d", &capacity);

    printf("Enter weight of %d items:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &weight[i]);
    }

    // Sort items in descending order (important optimization)
    qsort(weight, n, sizeof(int), compare);

    get_initial_upper_bound();

    for (int i = 0; i < n; i++)
        bin_load[i] = 0;

    solve(0, 0);
```

Output

Clear

```
--- Bin Packing: Branch and Bound ---
Enter number of items: 8
Enter bin capacity: 9
Enter weight of 8 items:

=== Session Ended. Please Run the code again ===
```

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```c

    printf("Enter number of items: ");
    scanf("%d", &n);

    printf("Enter bin capacity: ");
    scanf("%d", &capacity);

    printf("Enter weight of %d items:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &weight[i]);
    }

    // Sort items in descending order (important optimization)
    qsort(weight, n, sizeof(int), compare);

    get_initial_upper_bound();

    for (int i = 0; i < n; i++)
        bin_load[i] = 0;

    solve(0, 0);

    printf("\nMinimum bins required: %d\n", min_bins);

    return 0;
}
```

Output    Clear

```
--- Bin Packing: Branch and Bound ---
Enter number of items: 8
Enter bin capacity: 9
Enter weight of 8 items:

=== Session Ended. Please Run the code again ===
```